

AI Model Architecture Interview Guide

1. Transformer Model Families

Transformer models are broadly divided into:

- Encoder Only (BERT, DistilBERT, RoBERTa)
- Encoder-Decoder (T5, FLAN-T5, BART)
- Decoder Only (GPT, Qwen, Llama, Mistral, Gemma)

2. Encoder-Only Models (DistilBERT)

Purpose: Understanding text.

Common use cases:

- Intent Classification
- Sentiment Analysis
- NER

Architecture:

Input → Encoder → Classification Head → Prediction

Your project: DistilBERT + GLNER for intent classification and entity extraction.

3. Encoder-Decoder Models (FLAN-T5)

Purpose: Transform one text into another.

Common use cases:

- Text-to-SQL
- Translation
- Summarization

Architecture:

Input → Encoder → Context → Decoder → Output

Your project: FLAN-T5 SQL Generator.

4. Decoder-Only Models (Qwen/GPT/Llama)

Purpose: Text generation.

Architecture:

Input Tokens → Decoder Layers → Next Token Prediction

Common use cases:

- Chatbots
- RAG
- Agents
- Structured Extraction

Your project: Qwen 2.5 JSON Extraction.

5. LoRA

LoRA (Low Rank Adaptation):

- Freeze base model.
- Train only small adapter layers.

Benefits:

- Lower cost
- Faster training
- Smaller model artifacts

6. QLoRA

QLoRA combines LoRA with 4-bit quantization.

Benefits:

- Much lower GPU memory usage
- Can train larger models on T4/A10G GPUs

7. Interview Questions

Q: Why DistilBERT?

A: Fast encoder model for text understanding tasks.

Q: Why FLAN-T5?

A: Excellent for structured text transformation tasks.

Q: Why Qwen?

A: Strong instruction-following decoder-only LLM suitable for chat and extraction.

8. Your Project Mapping

DistilBERT + GLiNER → Intent Classification + Entity Extraction

FLAN-T5 + LoRA → Text-to-SQL Generation

Qwen + QLoRA → Structured JSON Extraction

Together these demonstrate experience with all three major transformer architectures.